Conduit Test plan

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Introduction

The “Conduit” application is designed as an open-source application to train software development and testing skills. The main website functionality includes registration and sign-in, posting articles on different topics, reading articles, the ability to like an article, posting a comment, and following users.
The Test Plan is designed to prescribe the scope, approach, resources, and schedule of all testing activities of the “Conduit” website.
The plan identifies the features to be tested, the types of testing to be performed, the resources and schedule required to complete testing, and the risks associated with the plan.

1. Test strategy

1.1 Testing scope

1.1.1 Features to be tested

Module name	Login type	Description
Conduit logo	Logged in Logged out	The logo is situated on all the pages. Clicking on this button opens the Home page.
Home	Logged in Logged out	On the Home tab, the user can see all published articles listed. Articles are sorted from newest to oldest. One page contains 10 articles. Users can sort articles by clicking on the necessary tag from the “Popular tags”. Click on [Home] opens the “Home page”
Sign in	Logged out	“Sign in” which allows the registered user to open a personal account. Users can log in with the filled "Email" and "Password" fields. If the user doesn’t have an account, it is possible to click on [Need an account?]

Sign up	Logged out	“Sign up” allows the user to create an account. Users can sign up with filled “Username”, "Email" and "Password" fields with a unique username and email.
Global Feed	Logged out	On this tab, users can see articles by authors. Each article has “Username”, “Title”, “Text area”, likes, and tags. The user can open the article by clicking on [Username], [Title], [Text area], and [Read more…] and read it. The opened article consists of “Profile Picture”, “Username”, “Follow”, “My Posts”, and “Favorited Posts”, which redirect to the “Sign in” page after clicking on each of them.
	Logged in	On this tab, users can see articles by authors. Each article has a “Username”, “Title”, “Text area”, likes, and tags. The user can open the article by clicking on [Username], [Title], [Text area], and [Read more…] and read it. The opened article consists of “Profile Picture”, “Username”, “Follow”, “My Posts”, and “Favorited Posts”. Users can [Follow] the author, like the author, and post a comment to the article.
Your Feed	Logged in	On this tab, users can see articles of the following authors. After clicking on [Follow], posts of this author appear in Your Feed.
Settings	Logged in	In Settings, the user can add a profile picture, change the username, write a short bio, and change email and password. After clicking on [Update settings], profile information changes and saves. After clicking on [Click here to log out] user logs out.
Profile	Logged in	The profile contains “My posts” with my articles and

		“Favorited posts” with liked articles. Articles are sorted from newest to oldest. One page contains 10 newest articles.
Popular tags	Logged in Logged out	In Popular tags, users can see the most popular tags. Click on Tag's title to open a list of articles that have this tag.

1.1.2 Out of test scope

These areas are out of scope for this testing cycle:

- Website Security
- Website Performance
- Website API testing
- Test Automation

1.2 Test types

In this testing cycle, the Conduit website will be tested using the next testing types:

- System testing: all the testing will be conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements.
    - Exploratory testing.
    - Smoke testing for each new build.
    - Functional testing of all the features
    - GUI testing
    - Compatibility testing: test only Windows and MacOS browsers

1.3 Risks and Issues

Risk	Mitigation
Team members lack the required skills for	Attend all the lectures in MA and do homework

website testing	on time
Not enough time to test all browsers and OS.	Create a pairwise table and split the tests according to it.
Not enough time to execute all test scenarios.	Prioritize tests and execute in the priority order.
A member of the team has got sick	Allocate time for emergencies in the testing schedule

1.4 Test Logistics

1.4.1 Who will test?

QA team will work on Test Cases creation and execution.

1.4.2 When will the testing occur?

The team will start the testing after:

- Creating decomposition, decision table, and state transition diagram.
- Creating accounts for Jira and TestRail.
- Test data and test cases were created.
- The software is available for testing.

2. Test objective

The test objectives are to verify the functionality and API of the Conduit app. The testing should be focused on the flow of publishing articles and sharing information between members. The main features are authorization, posting new articles, following members, saving favorite articles on their pages, adding likes, and writing comments. Testing should be done on preselected versions of browsers and mobile devices described in the "System resource" section.

3. Test criteria

3.1 Suspension criteria

- 10% of P0/P1 tests failed, which should lead to the testing suspension until the development team fixes all the corresponding bugs.
- 30% of P2/P3 tests failed, which should lead to the testing suspension until the development team fixes all the corresponding bugs.

3. 2 Exit criteria

The test execution will be stopped no later than the last day of the sprint. The exit criteria should be met to complete the testing cycle:

- The mandatory Execution rate is 95%.
- The mandatory Pass rate is 100% for P0/P1 tests.
- The mandatory Pass rate is 80% for P2/P3 tests.
- All necessary artifacts collected: test cases, and bug reports.
- The product should not have known bugs with severity Critical and Major and bugs with Priority High at the time of finishing testing.
- The list of allowed bugs is agreed upon with the developers and managers.

4. Resource planning

4.1 System resources

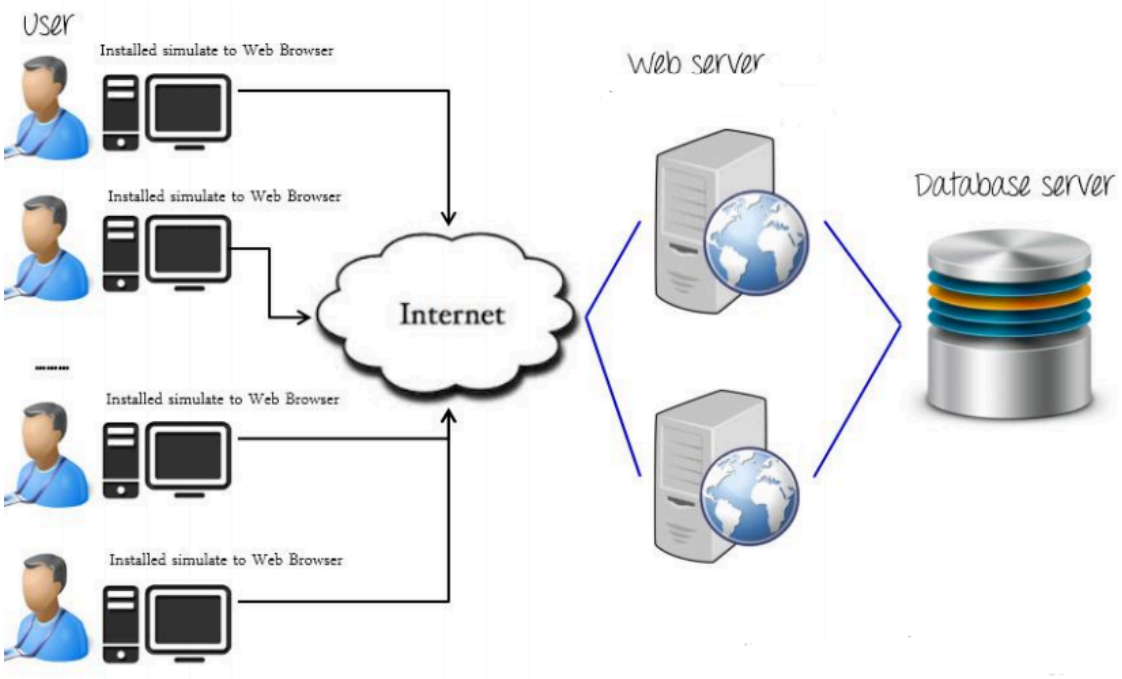
No.	Resources	Description
1	Browser	Google Chrome, Safary
2	Network	Stable Wi-Fi or Ethernet connection with a speed of at least 20 Mbps
3	Computer	Windows 11, macOS Monterey

4.2 Human resources

No.	**Resources**	**Description of tasks**
1	QA members	• Create decomposition; decision table; and state transition diagram • Create Test cases, Bug reports • Execute Test cases • Create Test report
2	Mentors	• Review test cases.

5. Test environment

Testing should be conducted in the production environment. To run the app locally for working with DB we will use Docker.



6. Schedule & estimation

6.1 All project tasks and estimation

Task	Members	Estimate effort
Create Test plan	QA members	3 man-hour
Create decomposition, decision table, state transition diagram	QA members	15 man-hour
Create Test cases	QA members	20 man-hour
Review Test cases	Mentors	3 man-hour
Test cases execution	QA members	10 man-hour
Create Bug reports	QA members	6 man-hour
Writing test report	QA members	6 man-hour

6.2 Schedule to complete these tasks

Task	1-st Sprint	2-nd Sprint	3-rd Sprint
Create Test plan	☑		
Create decomposition, decision table, state transition diagram	☑		
Create Test cases		☑	
Review Test cases		☑	
Test cases execution			☑
Create Bug reports			☑
Writing and preparing test results			☑

7. Test deliverables

7.1 Before the testing phase

- Test Plan document.
- Test Cases.
- Decomposition
- Decision table
- State transition diagram.

7.2 During the testing

- Test Cases with execution status.
- Bug reports.

7.3 After the testing cycle is over

- Test Report